Part J2 Energy efficiency

NT Part J2

Introduction to this Part

This Part sets out the application of the *Deemed-to-Satisfy Provisions* in Parts J3 to J9.

Notes

From 1 May 2023 to 30 September 2023 Section J of NCC 2019 Volume One Amendment 1 may apply instead of Section J of NCC 2022 Volume One. From 1 October 2023 Section J of NCC 2022 Volume One applies.

Notes: New South Wales Section J Energy Efficiency

(1) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Single Dwelling or Multi Dwelling Certificate issued under Version 3.0 or earlier, NSW Section J of NCC 2019 Volume One Amendment 1 applies.

(2) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Single Dwelling or Multi Dwelling Certificate issued under Version 4.0 or later, Section J of NCC 2022 Volume One applies.

(3) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Alterations and Additions Certificate, NSW Section J of NCC 2019 Volume One Amendment 1 applies.

(4) For a Class 3 building or Class 5 to 9 building:

(a) From 1 May 2023 to 30 September 2023 NSW Section J of NCC 2019 Volume One Amendment 1 may apply instead of Section J of NCC 2022 Volume One.

(b) From 1 October 2023 Section J of NCC 2022 Volume One applies.

Notes: Tasmania Section J Energy Efficiency

In Tasmania, for a Class 2 building and Class 4 part of a building, Section J is replaced with Section J of BCA 2019 Amendment 1.

Deemed-to-Satisfy Provisions

J2D1 Deemed-to-Satisfy Provisions

[2019: J0.0]

NSW J2D1(1)

(1) Where a *Deemed-to-Satisfy Solution* is proposed, *Performance Requirements* J1P1 to J1P4 are satisfied by complying with—

(a) J2D2; and

(b) J3D2 to J3D15; and

(c) J4D2 to J4D7; and

(d) J5D2 to J5D8; and

(e) J6D2 to J6D13; and

(f) J7D2 to J7D9; and

(g) J8D2 to J8D4; and

(h) J9D2 to J9D5.

(2) Where a *Performance Solution* is proposed, the relevant *Performance Requirements* must be determined in accordance with A2G2(3) and A2G4(3) as applicable.

NSW J2D2

J2D2 Application of Section J

[2019: J0.1]

(1) For a Class 2 to 9 building, other than a sole-occupancy unit of a Class 2 building or a Class 4 part of a building, *Performance Requirement* J1P1 is satisfied by complying with—

(a) Part J4, for the building *fabric*; and

(b) Part J5, for building sealing; and

(c) Part J6, for *air-conditioning* and ventilation; and

(d) Part J7, for artificial lighting and power; and

(e) Part J8, for heated water supply and *swimming pool* and spa pool plant; and

(f) J9D3, for facilities for energy monitoring.

(2) For a *sole-occupancy unit* of a Class 2 building or a Class 4 part of a building, *Performance Requirement* J1P2 is satisfied by complying with—

(a) J3D3, using *house energy rating software*; or

(b) the following—

(i) J3D4, for ceiling fans; and

(ii) J3D5, J3D6, J4D3, J4D7(3), J4D7(4) and Part J5, for general thermal construction; and

(iii) J3D7, for roofs; and

(iv) J3D8 and J3D11 to J3D13, or J3D9, for walls and *glazing*; and

(v) J3D10, for floors.

(3) For a *sole-occupancy unit* of a Class 2 building or a Class 4 part of a building, *Performance Requirement* J1P3 is satisfied by complying with—

(a) for the net equivalent energy usage—

(i) J3D14, for a *sole-occupancy unit* of a Class 2 building or a Class 4 part of a building with a total floor area not greater than 500 m^2; or

(ii) J3D15, using *house energy rating software*; and

(b) Part J6, for *air-conditioning* and ventilation; and

(c) Part J7, for artificial lighting and power.

(4) For a Class 2 to 9 building, *Performance Requirement* J1P4 is satisfied by complying with J9D4 and J9D5.

Part J3 Elemental provisions for a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

NT Part J3

TAS Part J3

Introduction to this Part

This Part contains *Deemed-to-Satisfy Provisions* (elemental) for compliance with Part J1. It sets out provisions for the insulation of building *fabric* and the energy efficiency of *domestic services* of a *sole-occupancy unit* of a Class 2 building or a Class 4 part of a building.

Notes

From 1 May 2023 to 30 September 2023 Section J of NCC 2019 Volume One Amendment 1 may apply instead of Section J of NCC 2022 Volume One. From 1 October 2023 Section J of NCC 2022 Volume One applies.

Notes: New South Wales Section J Energy Efficiency

(1) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Single Dwelling or Multi Dwelling Certificate issued under Version 3.0 or earlier, NSW Section J of NCC 2019 Volume One Amendment 1 applies.

(2) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Single Dwelling or Multi Dwelling Certificate issued under Version 4.0 or later, Section J of NCC 2022 Volume One applies.

(3) For a Class 2 building or a Class 4 part of a building, where a relevant *development consent* or an application for a complying development certificate requires compliance with a BASIX Alterations and Additions Certificate, NSW Section J of NCC 2019 Volume One Amendment 1 applies.

Deemed-to-Satisfy Provisions

J3D1 Deemed-to-Satisfy Provisions

[New for 2022]

NSW J3D1(1)

(1) Where a *Deemed-to-Satisfy Solution* is proposed, *Performance Requirements* J1P1 to J1P4 are satisfied by complying with—

(a) J2D2; and

(b) J3D2 to J3D15; and

(c) J4D2 to J4D7; and

(d) J5D2 to J5D8; and

(e) J6D2 to J6D13; and

(f) J7D2 to J7D9; and

(g) J8D2 to J8D4; and

(h) J9D2 to J9D5.

(2) Where a *Performance Solution* is proposed, the relevant *Performance Requirements* must be determined in accordance with A2G2(3) and A2G4(3) as applicable.

NSW J3D2

J3D2 Application of Part

[New for 2022]

The *Deemed-to-Satisfy Provisions* of this Part apply to building elements forming the external building *fabric* and *domestic services* of a *sole-occupancy unit* of a Class 2 building and a Class 4 part of a building.

NSW J3D3

J3D3 Reducing heating and cooling loads of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building using house energy rating software

[2019: J0.2]

(1) The *sole-occupancy units* of a Class 2 building or a Class 4 part of a building must—

(a) for reducing the heating or cooling loads—

(i) collectively achieve an average energy rating of not less than 7 stars, including the separate heating and cooling load limits; and

(ii) individually achieve an energy rating of not less than 6 stars, including the separate heating and cooling load limits; and

(b) for thermal breaks, comply with J3D5 and J3D6; and

(c) for compensating for a loss of ceiling insulation, other than where the *house energy rating software* has compensated for a loss of ceiling insulation, comply with Table J3D7w; and

(d) for general thermal construction, comply with J4D3; and

(e) for floor edge insulation, comply with J3D10(3), J3D10(5) and J3D10(6); and

(f) for building sealing, comply with Part J5.

(2) Energy ratings referred to in (1)(a)(i) and (ii) must be achieved using—

(a) *house energy rating software*; and

(b) the load limits specified in the ABCB Standard for NatHERS Heating and Cooling Load Limits.

NSW J3D4

J3D4 Ceiling fans in a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[2019: J0.3]

(1) Ceiling fans must be installed in accordance with Table J3D4 in—

(a) *climate zones* 1, 2 and 3; and

(b) *climate zone* 5 in New South Wales and Queensland.

(2) Ceiling fans *required* by (1) must—

(a) be permanently installed; and

(b) have a speed controller.

Table J3D4: Minimum ceiling fan requirements in climate zones 1, 2, 3 and 5

Size of room (m^2)	Minimum number and diameter (mm) of ceiling fans *required* for a bedroom in *climate zones* 1, 2 and 3	Minimum number and diameter (mm) of ceiling fans *required* in a *habitable room* other than a bedroom in *climate zones* 1, 2, 3 and 5 (NSW and Qld)
< 15	1 x 900	1 x 900

Size of room (m^2)	Minimum number and diameter (mm) of ceiling fans *required* for a bedroom in *climate zones* 1, 2 and 3	Minimum number and diameter (mm) of ceiling fans *required* in a *habitable room* other than a bedroom in *climate zones* 1, 2, 3 and 5 (NSW and Qld)
≥ 15 to < 20	1 x 1200	1 x 1200
≥ 20 to < 25	1 x 1200	1 x 1400
≥ 25 to < 30	1 x 1400	2 x 1200
≥ 30 to < 45	1 x 1400	2 x 1400
≥ 45 to < 50	2 x 1400	3 x 1200
≥ 50	2 x 1400	3 x 1400

J3D5 Roof thermal breaks of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[2019: J0.4]

(1) A roof that—

(a) has metal sheet roofing directly fixed to metal purlins, metal rafters or metal battens; and

(b) does not have a ceiling lining or has a ceiling lining fixed directly to those metal purlins, metal rafters or metal battens,

must have a thermal break, consisting of a material with an *R-Value* of greater than or equal to R0.2, installed between the metal sheet roofing and its supporting metal purlins, metal rafters or metal battens.

(2) The requirements of (1) do not apply to roofs constructed using insulated sandwich panels.

J3D6 Wall thermal breaks of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[2019: J0.5]

(1) A metal-framed wall that forms part of the building *envelope* must have a thermal break, consisting of a material with an *R-Value* of not less than R0.2, installed at all points of contact between the external cladding and the metal frame if the wall—

(a) does not have a wall lining or has a wall lining that is fixed directly to the same metal frame; and

(b) is clad with weatherboards, fibre-cement or the like, or metal sheeting fixed to a metal frame.

(2) The requirements of (1) do not apply to walls constructed using insulated sandwich panels.

NSW J3D7

J3D7 Roofs and ceilings of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[New for 2022]

(1) Roof and ceiling insulation must achieve the minimum *R-Value*—

(a) in *climate zone* 1, in accordance with Tables J3D7a, J3D7b and J3D7c as applicable; and

(b) in *climate zone* 2, in accordance with Tables J3D7d, J3D7e and J3D7f as applicable; and

(c) in *climate zone* 3, in accordance with Tables J3D7g, J3D7h and J3D7i as applicable; and

(d) in *climate zone* 4, in accordance with Tables J3D7j, J3D7k and J3D7l as applicable; and

(e) in *climate zone* 5, in accordance with Tables J3D7m, J3D7n and J3D7o as applicable; and

(f) in *climate zone* 6—

(i) R3.5; or

(ii) if the roof contains *reflective insulation*, R3.0; and

(g) in *climate zones* 7 and 8, in accordance with Tables J3D7p, J3D7q and J3D7r as applicable.

(2) *Reflective insulation* installed to comply with (1) must—

(a) have a surface emittance of not more than 0.05; and

(b) be adjacent to a roof space of not less than 20 mm; and

(c) in *climate zones* 3 to 8, be downward facing.

(3) The thermal bridging in a metal-framed roof must be addressed as follows—

(a) for a pitched roof with a horizontal ceiling—

(i) achieving the *Total R-Value* in Table J3D7s, calculated using a method that accounts for the effects of thermal bridging; or

(ii) increasing the *R-Value* of the insulation between the ceiling frames by R0.5 more than the *R-Value* derived from (1); or

(iii) adding a continuous ceiling insulation layer with a minimum *R-Value* of R0.13 above or below the ceiling joists or the bottom chords of the trusses; or

(iv) achieving the *required* ceiling *R-Value* derived from (1) by stacking two layers of insulation immediately on top of each other, such that the top layer is orientated to cover the ceiling joists or bottom chord of the trusses and has an *R-Value* of at least R0.5; or

(b) for a flat, skillion or cathedral roof—

(i) achieving the *Total R-Value* in Table J3D7t, calculated using a method that accounts for the effects of thermal bridging; or

(ii) complying with Table J3D7u.

(4) Where F8D5(1) applies, continuous insulation placed above the *primary insulation layer* to mitigate thermal bridging must have a *vapour permeance* of not less than that of the *primary insulation layer*.

(5) Where, for operational or safety reasons, the area of ceiling insulation *required* is reduced, the loss of insulation must be compensated for in accordance with Table J3D7w.

(6) Where the ceiling insulation *required* by (1) to (5) has an *R-Value*—

(a) greater than R3.0 and less than or equal to R4.5, it may be reduced to R3.0 within 450 mm of an *external wall*; or

(b) greater than R4.5, it may be reduced to R3.0 within 450 mm of an *external wall*, provided all other *required* ceiling insulation is increased by R0.5.

(7) The requirements of (1) to (6) do not apply to roofs constructed using insulated sandwich panels.

(8) Roofs constructed using insulated sandwich panels must achieve the minimum *Total R-Value* in Table J3D7x.

(9) In *climate zones* 1 to 5, the solar absorptance of the upper surface of a roof must not be more than 0.64.

Table J3D7a: Flat concrete roof — minimum R-Value for ceiling insulation: climate zone 1

Reflective insulation under-roof	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64
Yes	1.0	1.5	2.0	2.5	3.0
No	1.5	2.0	2.5	3.0	4.0

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(3) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7b: Timber-framed pitched roof with horizontal ceiling — minimum R-Value for ceiling insulation: climate zone 1

Roof ventilation	*Reflective insulation* under-roof	Under-roof insulation *R-Value*	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64
Vented	Yes	< 1.0	1.5	1.5	1.5	1.5	2.0
		≥ 1.0 to < 1.5	1.5	1.5	1.5	1.5	1.5
		≥ 1.5 to < 2.0	1.5	1.5	1.5	1.5	1.5
		≥ 2.0	1.5	1.5	1.5	2.0	2.0
	No	< 1.0	2.0	2.5	4.0	5.0	X
		≥ 1.0 to < 1.5	1.5	1.5	1.5	1.5	2.5
		≥ 1.5 to < 2.0	1.5	1.5	1.5	2.0	2.0
		≥ 2.0	1.5	1.5	1.5	1.5	1.5
Standard	Yes	< 1.0	1.5	1.5	2.0	3.0	4.0
		≥ 1.0 to < 1.5	1.5	1.5	1.5	1.5	2.5
		≥ 1.5 to < 2.0	1.5	1.5	1.5	1.5	2.0
		≥ 2.0	1.5	1.5	1.5	1.5	1.5
	No	< 1.0	2.5	4.0	6.0	X	X
		≥ 1.0 to < 1.5	1.5	1.5	1.5	1.5	4.0
		≥ 1.5 to < 2.0	1.5	1.5	2.0	2.0	2.5
		≥ 2.0	1.5	1.5	1.5	2.0	2.5

Table Notes

(1) SA = solar absorptance.

(2) A roof space is to be considered 'vented' if it—

(a) has one wind-driven roof ventilator per 50 m^2 of ceiling area with gable, eave or ridge vents; or

(b) has one powered roof ventilator per 200 m^2 of ceiling area with gable, eave or ridge vents; or

(c) is a tiled roof without *sarking-type material* at roof level.

(3) If a roof is not 'vented', it is a 'standard' roof.

(4) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(5) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

(6) X = not permitted.

Table J3D7c: Timber-framed flat, skillion or cathedral roof — minimum R-Value for ceiling insulation: climate zone 1

Reflective insulation under-roof	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64
Yes	1.5	1.5	2.0	3.0	4.0
No	1.5	3.5	5.0	X	X

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* can be achieved by installing insulation under the roof or on top of the ceiling or a combination of both.

(3) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(4) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

(5) X = not permitted.

Table J3D7d: Flat concrete roof — minimum R-Value for ceiling insulation: climate zone 2

Reflective insulation under-roof	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64
Yes	2.0	2.5	2.5	3.0	3.5
No	3.0	3.0	3.5	4.0	4.5

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(3) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7e: Timber-framed pitched roof with horizontal ceiling — minimum R-Value for ceiling insulation: climate zone 2

Roof ventilation	*Reflective insulation* under-roof	Under-roof insulation *R-Value*	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64
Vented	Yes	Any	2.5				
	No	0 to < 0.5	2.5	3.0	3.0	3.5	3.5
		≥ 0.5	2.5				
Standard	Yes	Any	2.5				
	No	0 to < 0.5	3.0	3.0	3.5	4.0	4.0
		≥ 0.5 to < 1.0	2.5	2.5	2.5	3.0	3.0
		≥ 1.0	2.5				

Table Notes

(1) SA = solar absorptance.

(2) A roof is considered 'vented' if it—

(a) has one wind-driven roof ventilator per 50 m^2 of ceiling area with gable, eave or ridge vents; or

(b) has one powered roof ventilator per 200 m^2 of ceiling area with gable, eave or ridge vents; or

(c) is a tiled roof without *sarking-type material* at roof level.

(3) If a roof is not 'vented', it is a 'standard' roof.

(4) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(5) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7f: Timber-framed flat, skillion or cathedral roof — minimum R-Value for ceiling insulation: climate zone 2

Reflective insulation under-roof	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64
Yes	2.5	2.5	2.5	2.5	2.5
No	3.0	3.0	3.5	4.0	4.0

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* can be achieved by installing insulation under the roof or on top of the ceiling or a combination of

both.

(3) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(4) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7g: Flat concrete roof — minimum R-Value for ceiling insulation: climate zone 3

Reflective insulation under-roof	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64
Yes	2.0	2.0	2.5	2.5	3.0
No	3.0	3.5	4.0	4.5	5.0

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(3) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7h: Timber-framed pitched roof with horizontal ceiling — minimum R-Values for ceiling insulation: climate zone 3

Roof ventilation	*Reflective insulation* under-roof	Under-roof insulation *R-Value*	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64
Vented	Yes	< 0.5	2.5	2.5	2.5	2.5	2.5
		≥ 0.5 to < 1.0	2.0	2.0	2.0	2.0	2.0
		≥ 1.0 to < 1.5	2.5	2.0	2.0	2.0	2.0
		≥ 1.5 to < 2.0	2.0	2.0	2.5	2.5	2.5
		≥ 2.0	2.5	2.5	3.0	3.0	3.0
	No	< 0.5	3.5	4.0	4.5	5.0	X
		≥ 0.5 to < 1.0	3.0	3.5	3.5	4.0	4.5
		≥ 1.0 to < 1.5	2.5	3.0	3.0	3.0	3.5
		≥ 1.5 to < 2.0	2.5	3.0	3.0	3.0	3.0
		≥ 2.0	2.5	2.5	3.0	3.0	3.0
Standard	Yes	< 0.5	2.0	2.0	2.5	2.5	2.5
		≥ 0.5 to < 1.0	2.0	2.0	2.5	2.5	2.5
		≥ 1.0 to < 1.5	2.0	2.0	2.0	2.0	2.5
		≥ 1.5 to < 2.0	2.0	2.5	2.5	2.5	3.0
		≥ 2.0	2.0	2.0	2.5	2.5	2.5
	No	< 0.5	3.5	4.0	5.0	X	X
		≥ 0.5 to < 1.0	3.0	3.0	3.5	4.0	5.0
		≥ 1.0 to < 1.5	2.5	2.5	3.0	3.0	3.5
		≥ 1.5 to < 2.0	2.0	2.0	2.5	2.5	2.5
		≥ 2.0	2.0	2.0	2.5	2.5	2.5

Table Notes

(1) SA = solar absorptance.

(2) A roof is considered 'vented' if it—

(a) has one wind-drive roof ventilator per 50 m^2 of ceiling area with gable, eave or ridge vents; or

(b) has one powered roof ventilator per 200 m^2 of ceiling area with gable, eave or ridge vents; or

(c) is a tiled roof without *sarking-type material* at roof level.

(3) If a roof is not 'vented', it is a 'standard' roof.

(4) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(5) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

(6) X = not permitted.

Table J3D7i: Timber-framed flat, skillion or cathedral roof — minimum R-Value for ceiling insulation: climate zone 3

Reflective insulation under-roof	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64
Yes	2.0	2.0	2.5	2.5	2.5
No	3.5	4.0	5.0	X	X

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* can be achieved by installing insulation under the roof or on top of the ceiling or a combination of both.

(3) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(4) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

(5) X = not permitted.

Table J3D7j: Flat concrete roof — minimum R-Value for ceiling insulation: climate zone 4

Reflective insulation under-roof	≤ 0.64
Yes	2.0
No	2.5

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(3) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7k: Timber-framed pitched roof with horizontal ceiling — minimum R-Value for ceiling insulation: climate zone 4

Roof ventilation	*Reflective insulation* under- roof	Under-roof insulation *R-Value*	SA ≤ 0.23	SA > 0.23 to ≤ 0.64
Vented	Yes	< 0.5	3.0	3.5
		≥ 0.5	3.0	3.0
	No	Any	3.5	
Standard	Yes	Any	3.0	
	No	< 0.5	3.5	
		≥ 0.5	3.0	

Table Notes

(1) SA = solar absorptance.

(2) A roof is considered 'vented' if it—

(a) has one wind-driven roof ventilator per 50 m^2 of ceiling area with gable, eave or ridge vents; or

(b) has one powered roof ventilator per 200 m^2 of ceiling area with gable, eave or ridge vents; or

(c) is a tiled roof without *sarking-type material* at roof level.

(3) If a roof is not 'vented', it is a 'standard' roof.

(4) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(5) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7l: Timber-framed flat, skillion or cathedral timber roof — minimum R-Value for ceiling insulation: climate zone 4

Reflective insulation under-roof	SA ≤ 0.64
Yes	3.0
No	3.5

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* can be achieved by installing insulation under the roof or on top of the ceiling or a combination of both.

(3) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(4) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7m: Flat concrete roof — minimum R-Value for ceiling insulation: climate zone 5

Reflective insulation under-roof	SA ≤ 0.42	SA > 0.42 to ≤ 0.64
Yes	3.0	3.5
No	4.0	4.0

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(3) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7n: Timber-framed pitched roof with horizontal ceiling — minimum R-Value for ceiling insulation: climate zone 5

Roof ventilation	*Reflective insulation* under- roof	Under-roof insulation *R-Value*	SA ≤ 0.42	SA > 0.42 to ≤ 0.64
Vented	Yes	< 0.5	3.0	2.5
		≥ 0.5	2.5	
	No	0 to < 2.0	3.0	
		≥ 2.0	2.5	
Standard	Yes	Any	2.5	
	No	0 to < 1.0	3.0	
		≥ 1.0 to < 2.0	2.5	
		≥ 2.0	3.0	

Table Notes

(1) SA = solar absorptance.

(2) A roof is considered 'vented' if it—

(a) has one wind-driven roof ventilator per 50 m^2 of ceiling area with gable, eave or ridge vents; or

(b) has one powered roof ventilator per 200 m^2 of ceiling area with gable, eave or ridge vents; or

(c) is a tiled roof without *sarking-type material* at roof level.

(3) If a roof is not 'vented', it is a 'standard' roof.

(4) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(5) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7o: Timber-framed flat, skillion or cathedral roof — minimum R-Value for ceiling insulation: climate zone 5

Reflective insulation under- roof	*R-Value*
Yes	2.5
No	3.0

Table Notes

(1) The *R-Value* can be achieved by installing insulation under the roof or on top of the ceiling or a combination of both.

(2) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(3) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7p: Flat concrete roof — minimum R-Value for ceiling insulation: climate zones 7 and 8

Reflective insulation under-roof	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64	SA > 0.64 to ≤ 0.73	SA > 0.73
Yes	3.5	3.5	3.5	3.0	3.0	3.0	2.5
No	4.0	3.0	3.0	2.5	2.5	2.5	2.0

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(3) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7q: Timber-framed pitched roof with flat ceiling — minimum R-Value for ceiling insulation: climate zones 7 and 8

Roof ventilation	*Reflective insulation* under-roof	Under-roof insulation *R-Value*	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64	SA > 0.64 to ≤ 0.73	SA > 0.73 to ≤ 0.85	SA > 0.85
Standard	Yes	0 to < 1.0	4.5	4.5	4.5	4.0	4.0	4.0	4.0	4.0
		≥ 1.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
	No	0 to < 1.0	5.0	4.5	4.5	4.0	4.0	3.5	3.5	3.5
		≥ 1.0 to < 1.5	4.5	4.5	4.0	4.0	4.0	4.0	3.5	3.5
		≥ 1.5	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
Vented	Yes	0 to < 1.0	4.5	4.5	4.5	4.5	4.5	4.0	4.0	4.0
		≥ 1.0	4.5	4.5	4.5	4.5	4.0	4.0	4.0	4.0
	No	0 to < 1.0	5.0	4.5	4.5	4.5	4.0	4.0	3.5	3.5
		≥ 1.0 to < 1.5	4.5	4.5	4.5	4.5	4.0	4.0	4.0	4.0
		≥ 1.5	4.5	4.5	4.5	4.5	4.5	4.0	4.0	4.0

Table Notes

(1) SA = solar absorptance.

(2) A roof is considered 'vented' if it—

(a) has one wind-driven roof ventilator per 50 m^2 of ceiling area with gable, eave or ridge vents; or

(b) has one powered roof ventilator per 200 m^2 of ceiling area with gable, eave or ridge vents; or

(c) is ventilated to *outdoor air* through evenly distributed openings in accordance with Table F8D5; or

(d) is a tiled roof without *sarking-type material* at roof level.

(3) If a roof is not 'vented', it is a 'standard' roof.

(4) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(5) *R-Values* listed are for the labelled, declared *R-Value* of insulation.

Table J3D7r: Timber-framed flat, skillion or cathedral roof — minimum R-Value for ceiling insulation: climate zones 7 and 8

Reflective insulation under-roof	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64	SA > 0.64 to ≤ 0.73	SA > 0.73 to ≤ 0.85	SA > 0.85
Yes	4.5	4.5	4.5	4.0	4.0	4.0	4.0	4.0
No	5.0	4.5	4.5	4.0	4.0	3.5	3.5	3.5

Table Notes

(1) SA = solar absorptance.

(2) The *R-Value* can be achieved by installing insulation under the roof or on top of the ceiling or a combination of both.

(3) The *R-Value* of *reflective insulation* is not to be included in the *R-Value* of any under-roof or ceiling insulation.

(4) *R-Values* listed are for the labelled, declared *R-Value* for insulation.

Table J3D7s: Metal-framed pitched roof with horizontal ceiling — minimum Total R-Value of ceiling to account for thermal bridging

Ceiling insulation *R-Value* from Tables J3D7a to J3D7r and J3D7(1)(f) as applicable	Minimum *Total R-Value* to account for thermal bridging
1.5	1.38
2.0	1.74
2.5	2.09
3.0	2.43
3.5	2.63
4.0	2.95
4.5	3.27
5.0	3.59
5.5	3.91
6.0	4.23

Table Notes

(1) The *Total R-Value* calculation only includes the ceiling frame, insulation and ceiling lining. It is not to include the internal air films, roof space or roof lining.

(2) Minimum ceiling *Total R-Values* are in-situ values. They account for compression of insulation.

Table J3D7t: Metal-framed flat, skillion or cathedral roof — minimum Total R-Value to account for thermal bridging

Ceiling insulation *R-Value* from Tables J3D7a to J3D7r, and J3D7(1)(f) as applicable	Minimum *Total R-Value* to account for thermal bridging: heat flow down	Minimum *Total R-Value* to account for thermal bridging: heat flow up
1.0	1.40	1.32
1.5	1.86	1.78
2.0	2.29	2.21
2.5	2.71	2.63
3.0	3.11	3.02
3.5	3.31	3.22
4.0	3.66	3.57
4.5	3.98	3.90
5.0	4.32	4.22
5.5	4.63	4.53
6.0	4.93	4.82

Table Notes

(1) Minimum *Total R-Values* are in-situ values. They account for compression of insulation.

(2) Direction of heat flow must be determined in accordance with Table J3D7v.

Table J3D7u: Metal-framed flat, skillion or cathedral roof – thermal bridging mitigation

Minimum ceiling *R-Value* from Tables J3D7a to J3D7r, and J3D7(1)(f) as applicable	Option 1– increase insulation between roof frame members to specified minimum *R-Value*	Option 2 – add a layer of continuous insulation with specified minimum *R-Value* above or below the roof frame members
1.0	1.5	0.13
1.5	2.5	0.30
2.0	3.5	0.30
2.5	5.0	0.40
3.0	6.0	0.60
3.5	X	0.60
4.0	X	0.60
4.5	X	0.60
5.0	X	0.60
5.5	X	0.60
6.0	X	0.60

Table Notes

(1) Minimum *R-Values* are in-situ values. They account for compression of insulation.

(2) X= not permitted.

Table J3D7v: Direction of heat flow

Climate zone	Direction of heat flow
1	Down
2 – altitude less than 300 m	Down
2 – altitude 300 m or more	Down and up

Climate zone	Direction of heat flow
3	Down and up
4	Up
5	Up
6	Up
7	Up
8	Up

Table J3D7w: Adjusted minimum R-Value of ceiling insulation required to compensate for loss of ceiling insulation area

Percentage of ceiling area uninsulated	Minimum *R-Value* of ceiling insulation *required* to satisfy J3D7(1) and J3D7(3)									
	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5
0.5% to less than 1.0%	1.0	1.6	2.2	2.8	3.4	4.0	4.7	5.4	6.2	6.9
1.0% to less than 1.5%	1.1	1.7	2.3	2.9	3.6	4.4	5.2	6.1	7.0	X
1.5% to less than 2.0%	1.1	1.7	2.4	3.1	3.9	4.8	5.8	6.8	X	X
2.0% to less than 2.5%	1.1	1.8	2.5	3.3	4.2	5.3	6.5	X	X	X
2.5% to less than 3.0%	1.2	1.9	2.6	3.6	4.6	5.9	X	X	X	X
3.0% to less than 4.0%	1.2	2.0	3.0	4.2	5.7	X	X	X	X	X
4.0% to less than 5.0%	1.3	2.2	3.4	5.0	X	X	X	X	X	X

Table Notes

(1) Interpolation is allowed for values between those shown.

(2) X = not permitted.

Table J3D7x: Total R-Values for roofs constructed with insulated sandwich panels

Climate zone	SA ≤ 0.23	SA > 0.23 to ≤ 0.32	SA > 0.32 to ≤ 0.42	SA > 0.42 to ≤ 0.53	SA > 0.53 to ≤ 0.64	SA > 0.64 to ≤ 0.73	SA > 0.73 to ≤ 0.85	SA > 0.85
1	1.86	3.31	4.32	X	X	X	X	X
2 – heat flow down	3.11	3.11	3.31	3.66	3.66	X	X	X
2 – heat flow up	3.02	3.02	3.22	3.57	3.57	X	X	X
3 – heat flow down	3.31	3.66	4.32	X	X	X	X	X
3 – heat flow up	3.22	3.57	4.22	X	X	X	X	X
4	3.22	3.22	3.22	3.22	3.22	X	X	X
5	3.02	3.02	3.02	3.02	3.02	X	X	X
6	3.57	3.57	3.57	3.57	3.57	3.57	3.57	3.57
7	4.22	3.90	3.90	3.57	3.57	3.22	3.22	3.22
8	3.90	3.57	3.57	3.22	3.22	3.02	3.02	3.02

Table Notes

(1) SA = solar absorptance.

(2) Direction of heat flow must be determined in accordance with Table J3D7v.

(3) X = not permitted.

NSW J3D8

J3D8 External walls of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[New for 2022]

(1) The *Total R-Value* of an *external wall*—

(a) in *climate zones* 1, 2, 3, 5 and 6—

(i) where the ratio of the area of opaque *external walls* to the *floor area* of the *sole-occupancy unit* is less than 20%, must be at least R1.15; and

(ii) where the ratio of the area of opaque *external walls* to the *floor area* of the *sole-occupancy unit* is greater than or equal to 20% but less than 35%, must be at least R2.04; and

(iii) where the ratio of the area of opaque *external walls* to the *floor area* of the *sole-occupancy unit* is greater than or equal to 35%, must be at least R2.24; and

(b) in *climate zones* 4, 7 and 8, must be at least R2.24.

(2) The *Total R-Value* of an *external wall* must be determined in accordance with—

(a) for a spandrel panel in a curtain wall system, in accordance with Specification 38; and

(b) for all other walls, in accordance with AS/NZS 4859.2.

(3) The solar absorptance of an *external wall* must—

(a) in *climate zones* 1 to 6, be in accordance with Table J3D8a; and

(b) in *climate zones* 7 and 8, be in accordance with Table J3D8b.

Table J3D8a: Solar absorptance – climate zones 1 to 6

Climate zone	Opaque *external wall* to net *floor area* ratio	Permitted solar absorptance
1 and 3	< 45%	≤ 0.8
	≥ 45%	≤ 0.35
2	< 35%	Any
	≥ 35%	Any, or ≤ 0.35, if shading device overhang is < 300 mm
4 and 5	< 45%	Any
	≥ 45%	≤ 0.35, if shading device overhang is < 1500 mm
6	Any	Any

Table J3D8b: Solar absorptance – climate zones 7 and 8

Opaque *external wall* to net *floor area* ratio	Shading device overhang (mm)			
	≥ 0 to < 600	≥ 600 to < 900	≥ 900 to < 1200	≥ 1200
< 20%	Any	Any	≥ 0.4	X
≥ 20% to < 35%	Any	Any	X	X
≥ 35% to < 45%	Any	≥ 0.4	X	X
≥ 45%	Any	≥ 0.6	X	X

Table Notes

X = not permitted

NSW J3D9

J3D9 Wall-glazing construction of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[New for 2022]

(1) The *Total System U-Value* of *wall-glazing construction* that forms part of the external building *fabric* must not be greater than—

(a) in *climate zones* 1 to 5, U2.2; or

(b) in *climate zone* 6, U2.0; or

(c) in *climate zones* 7 and 8, U1.4.

(2) The *Total System U-Value* of *wall-glazing construction* that forms part of the external building *fabric* must be calculated in accordance with Specification 37.

(3) Wall components of *wall-glazing construction* must achieve a minimum *Total R-Value* of—

(a) where the wall is less than 80% of the area of the *wall-glazing construction*, R1.0; or

(b) where the wall is 80% or more of the area of the *wall-glazing construction*, the value specified in Table J4D6a for a Class 3 building.

(4) In *climate zones* 1 to 6, the *solar admittance* of externally facing *wall-glazing construction* must be not greater than that shown in Table J3D9.

(5) In *climate zones* 7 and 8, *glazing* in a *wall-glazing construction* must have a *Total System SHGC* of at least 0.4.

(6) The *solar admittance* of a *wall-glazing construction* must be calculated in accordance with Specification 37.

(7) The solar absorptance of an *external wall* must be in accordance with J3D8(3).

Table J3D9: Maximum wall-glazing construction solar admittance

Climate zone	Eastern aspect *solar admittance*	Northern aspect *solar admittance*	Southern aspect *solar admittance*	Western aspect *solar admittance*
1	0.10	0.10	0.14	0.10
2	0.10	0.10	0.10	0.10
3	0.11	0.11	0.11	0.11
4	0.11	0.11	0.11	0.11
5	0.13	0.13	0.13	0.13
6	0.14	0.14	0.14	0.14
7	N/A	N/A	N/A	N/A
8	N/A	N/A	N/A	N/A

J3D10 Floors of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[New for 2022]

NSW J3D10(1)

(1) Where a *sole-occupancy unit* of a Class 2 building or a Class 4 part of a building has a concrete floor above an unenclosed *carpark*, undercroft, or the like, underfloor insulation must be installed with an *R-Value* at least—

(a) in *climate zone* 2 and *climate zones* 5 to 8, R2.0; and

(b) in *climate zones* 3 and 4, R1.5.

NSW J3D10(2)

(2) Where a *sole-occupancy unit* of a Class 2 building or Class 4 part of a building has a concrete floor above an enclosed *carpark*, undercroft or the like, underfloor insulation must be installed with an *R-Value* at least—

(a) in *climate zone* 2, R0.5; and

(b) in *climate zones* 4 and 5, R1.0; and

(c) in *climate zone* 6, R1.5; and

(d) in *climate zones* 7 and 8, R2.0.

(3) A concrete slab-on-ground with an in-slab or in-screed heating or cooling system must have insulation with an *R-Value* at least 1.0 installed around the vertical edge of tis perimeter.

NSW J3D10(4)

(4) Except for a waffle-pod slab—

(a) in *climate zones* 6 and 7—

(i) insulation with an *R-Value* of at least 0.64 must be installed around the vertical edge of its perimeter; and

(ii) insulation with an *R-Value* of at least 0.64 must be installed underneath the slab; and

(b) in *climate zone* 8—

(i) insulation with an *R-Value* of at least 1.0 must be installed around the vertical edge of its perimeter; and

(ii) insulation with an *R-Value* of at least 2.0 must be installed underneath the slab.

(5) Insulation *required* by (3), (4)(a)(i) and (4)(b)(i) must—

(a) be *water resistant*; and

(b) be continuous from the adjacent finished ground level—

(i) to a depth of not less than 300 mm; or

(ii) for at least the full depth of the vertical edge of the concrete slab-on-ground.

(6) The requirements of (3) do not apply to an in-screed heating or cooling system used solely in a bathroom, amenity area or the like.

NSW J3D11

J3D11 External winter glazing of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[New for 2022]

(1) In *climate zones* 2 to 8, the ratio of the conductance (C_U) and solar heat gain (C_{SHGC}) of the *glazing* of each *storey*, including any *mezzanine*, of a *sole-occupancy unit* of a Class 2 building or a Class 4 part of a building must—

(a) not exceed the allowance obtained from Table J3D11a; and

(b) be calculated in accordance with the following formula:

$$\frac{[(A_1 \times U_1 \times BC_1 \times OC_1 \times R_{W1}) + (A_2 \times U_2 \times BC_2 \times OC_2 \times R_{W2}) + \ldots]}{[(A_1 \times SHGC_1 \times E_{W1} \times BS_{W1} \times F_{W1} \times H_{W1} \times R_{W1}) + (A_2 \times SHGC_2 \times E_{W2} \times BS_{W2} \times F_{W2} \times H_{W2} \times R_{W2}) + \ldots]}$$

(2) In the formula at (1)(b)—

(a) $A_{1,2,etc}$ = the area of each *glazing* element; and

(b) $U_{1,2,etc}$ = the *Total System U-Value* of each *glazing* element; and

(c) $SHGC_{1,2,etc}$ = the *Total System SHGC* for each *glazing* element, not exceeding 0.7; and

(d) $E_{W1,2,etc}$ = the winter exposure factor for each *glazing* element obtained from Table J3D11b, J3D11c, J3D11d, J3D11e, J3D11f or J3D11g; and

(e) $BC_{1,2,etc}$ = the bedroom conductance factor obtained from Table J3D11h, J3D11i, J3D11j, J3D11k, J3D11l or J3D11m; and

(f) $OC_{1,2,etc}$ = the orientation sector conductance factor obtained from Table J3D11n; and

(g) $R_{W1,2,etc}$ = the room type factor obtained from Table J3D11h, J3D11i, J3D11j, J3D11k, J3D11l or J3D11m; and

(h) $BS_{W1,2,etc}$ = the bedroom solar heat gain factor obtained from Table J3D11h, J3D11i, J3D11j, J3D11k, J3D11l or J3D11m; and

(i) $F_{W1,2,etc}$ = the frame factor obtained from Table J3D11o, J3D11p, J3D11q, J3D11r, J3D11s or J3D11t for each *glazing* element; and

(j) $H_{W1,2,etc}$ = the floor factor obtained from Table J3D11h, J3D11i, J3D11j, J3D11k, J3D11l or J3D11m for each *glazing* element.

(3) For the purpose of J3D11—

(a) orientation sectors must be determined in accordance with Figure 13.3.2a of the ABCB Housing Provisions; and

(b) P/H must be determined in accordance with Figure S37C7; and

(c) For P/H between those in Tables J3D11b, J3D11c, J3D11d, J3D11e, J3D11f and J3D11g, either use the next highest P/H or interpolate.

Table J3D11a: Maximum conductance to solar heat gain ratio (C_U/C_{SHGC})

Climate zone	Maximum conductance to solar heat gain ratio (C_U/C_{SHGC})
2	16.95
3	19.88
4	13.34
5	11.83
6	6.27
7	12.90

Climate zone	Maximum conductance to solar heat gain ratio (C_U/C_{SHGC})
8	12.90

Table J3D11b: Orientation sector winter exposure factor (E_W): climate zone 2

P/H	North	North east	East	South east	South	South west	West	North west
0.00	1.49	1.61	1.23	0.96	0.68	1.03	1.37	1.71
0.05	1.44	1.53	1.14	0.81	0.57	0.90	1.27	1.64
0.10	1.38	1.48	1.09	0.76	0.53	0.85	1.22	1.55
0.20	1.21	1.32	0.97	0.69	0.50	0.75	1.10	1.39
0.40	1.00	1.06	0.77	0.57	0.42	0.63	0.92	1.14
0.60	0.83	0.87	0.69	0.50	0.37	0.55	0.77	0.96
0.80	0.62	0.69	0.56	0.43	0.35	0.50	0.66	0.79
1.00	0.43	0.59	0.46	0.41	0.33	0.43	0.56	0.65
1.20	0.26	0.47	0.40	0.36	0.31	0.40	0.53	0.53
1.40	0.22	0.40	0.33	0.33	0.29	0.38	0.47	0.44
1.60	0.18	0.32	0.31	0.33	0.28	0.35	0.39	0.39
1.80	0.15	0.28	0.27	0.31	0.26	0.33	0.35	0.32
2.00	0.12	0.21	0.24	0.29	0.26	0.33	0.33	0.31

Table J3D11c: Orientation sector winter exposure factor (E_W): climate zone 3

P/H	North	North east	East	South east	South	South west	West	North west
0.00	0.90	1.10	0.88	0.69	0.50	0.58	0.67	0.91
0.05	0.89	1.06	0.82	0.60	0.44	0.51	0.64	0.89
0.10	0.83	1.01	0.78	0.58	0.42	0.49	0.61	0.84
0.20	0.74	0.90	0.70	0.52	0.40	0.44	0.55	0.76
0.40	0.59	0.74	0.60	0.43	0.36	0.37	0.47	0.63
0.60	0.44	0.57	0.48	0.39	0.34	0.32	0.38	0.51
0.80	0.30	0.45	0.43	0.34	0.30	0.30	0.32	0.41
1.00	0.20	0.38	0.35	0.30	0.28	0.28	0.29	0.33
1.20	0.14	0.29	0.31	0.28	0.28	0.25	0.24	0.28
1.40	0.11	0.26	0.25	0.26	0.26	0.23	0.21	0.24
1.60	0.08	0.23	0.24	0.24	0.24	0.21	0.20	0.20
1.80	0.07	0.16	0.19	0.24	0.24	0.19	0.17	0.18
2.00	0.06	0.16	0.18	0.21	0.24	0.19	0.15	0.16

Table J3D11d: Orientation sector winter exposure factor (E_W): climate zone 4

P/H	North	North east	East	South east	South	South west	West	North west
0.00	1.58	1.66	1.16	0.93	0.70	0.78	0.85	1.38
0.05	1.54	1.60	1.06	0.79	0.58	0.66	0.79	1.32
0.10	1.53	1.54	1.04	0.74	0.56	0.62	0.75	1.28
0.20	1.30	1.41	0.94	0.67	0.50	0.56	0.68	1.15
0.40	1.18	1.20	0.78	0.57	0.44	0.48	0.58	0.99
0.60	0.98	0.99	0.69	0.50	0.38	0.42	0.49	0.81
0.80	0.85	0.81	0.60	0.45	0.36	0.38	0.44	0.68

P/H	North	North east	East	South east	South	South west	West	North west
1.00	0.68	0.73	0.52	0.41	0.32	0.34	0.37	0.58
1.20	0.49	0.56	0.46	0.36	0.32	0.32	0.34	0.50
1.40	0.38	0.52	0.42	0.33	0.30	0.30	0.30	0.42
1.60	0.27	0.45	0.39	0.33	0.28	0.28	0.26	0.37
1.80	0.21	0.39	0.35	0.31	0.28	0.26	0.24	0.32
2.00	0.19	0.35	0.31	0.29	0.26	0.24	0.22	0.26

Table J3D11e: Orientation sector winter exposure factor (E_W): climate zone 5

P/H	North	North east	East	South east	South	South west	West	North west
0.00	1.61	1.34	1.08	0.87	0.67	0.76	0.85	1.23
0.05	1.56	1.29	0.98	0.74	0.56	0.64	0.78	1.17
0.10	1.56	1.23	0.92	0.69	0.54	0.62	0.75	1.14
0.20	1.30	1.10	0.83	0.63	0.49	0.54	0.67	1.03
0.40	1.19	0.91	0.69	0.54	0.41	0.47	0.55	0.86
0.60	0.97	0.75	0.56	0.47	0.38	0.41	0.47	0.70
0.80	0.78	0.62	0.49	0.42	0.34	0.37	0.42	0.57
1.00	0.64	0.47	0.39	0.38	0.32	0.33	0.36	0.49
1.20	0.43	0.42	0.35	0.36	0.31	0.31	0.29	0.39
1.40	0.32	0.31	0.29	0.34	0.29	0.29	0.28	0.33
1.60	0.22	0.27	0.27	0.31	0.25	0.25	0.23	0.28
1.80	0.18	0.23	0.22	0.29	0.25	0.25	0.20	0.25
2.00	0.14	0.17	0.21	0.27	0.25	0.23	0.19	0.19

Table J3D11f: Orientation sector winter exposure factor (E_W): climate zone 6

P/H	North	North east	East	South east	South	South west	West	North west
0.00	3.04	2.50	1.52	1.51	1.51	1.63	1.76	2.75
0.05	2.94	2.36	1.39	1.28	1.26	1.38	1.62	2.61
0.10	2.91	2.28	1.33	1.21	1.19	1.30	1.52	2.55
0.20	2.50	2.05	1.18	1.08	1.05	1.16	1.40	2.34
0.40	2.29	1.77	1.01	0.91	0.91	0.98	1.20	1.98
0.60	1.95	1.51	0.86	0.77	0.81	0.83	1.04	1.71
0.80	1.73	1.28	0.72	0.71	0.74	0.80	0.92	1.42
1.00	1.38	1.02	0.65	0.64	0.67	0.69	0.78	1.24
1.20	1.12	0.95	0.55	0.61	0.60	0.65	0.72	1.04
1.40	0.85	0.72	0.49	0.54	0.60	0.62	0.64	0.93
1.60	0.70	0.65	0.42	0.50	0.56	0.54	0.56	0.79
1.80	0.51	0.53	0.40	0.47	0.53	0.51	0.54	0.72
2.00	0.40	0.47	0.38	0.44	0.49	0.51	0.48	0.58

Table J3D11g: Orientation sector winter exposure factor (E_W): climate zones 7 and 8

P/H	North	North east	East	South east	South	South west	West	North west
0.00	1.66	1.53	0.90	0.82	0.74	0.78	0.83	1.43
0.05	1.61	1.47	0.83	0.69	0.61	0.66	0.76	1.37

P/H	North	North east	East	South east	South	South west	West	North west
0.10	1.61	1.44	0.79	0.64	0.59	0.62	0.73	1.34
0.20	1.51	1.34	0.73	0.58	0.53	0.55	0.66	1.22
0.40	1.30	1.17	0.63	0.49	0.44	0.47	0.55	1.07
0.60	1.19	1.02	0.54	0.43	0.40	0.41	0.48	0.91
0.80	1.02	0.88	0.48	0.39	0.36	0.37	0.43	0.79
1.00	0.93	0.78	0.44	0.34	0.34	0.33	0.37	0.67
1.20	0.73	0.66	0.37	0.32	0.32	0.31	0.33	0.60
1.40	0.66	0.64	0.36	0.30	0.29	0.29	0.30	0.46
1.60	0.51	0.46	0.32	0.28	0.27	0.25	0.28	0.43
1.80	0.42	0.44	0.26	0.26	0.27	0.25	0.25	0.37
2.00	0.31	0.37	0.26	0.24	0.25	0.25	0.22	0.31

Table J3D11h: Conductance and radiation factors: climate zone 2

Type of factor	Factor
Bedroom conduction weighting factor (BC)	0.43
Room type multiplier (for bedroom and unconditioned areas) (R_W)	1.00
Bedroom solar heat gain weighting factor (BS_W)	0.40
Floor factor for tiled or vinyl covered floors (H_W)	1.02
Floor factor for other than tiled or vinyl covered floors (H_W)	0.96

Table J3D11i: Conductance and radiation factors: climate zone 3

Type of factor	Factor
Bedroom conduction weighting factor (BC)	0.42
Room type multiplier (for bedroom and unconditioned areas) (R_W)	1.00
Bedroom solar heat gain weighting factor (BS_W)	1.28
Floor factor for tiled or vinyl covered floors (H_W)	1.02
Floor factor for other than tiled or vinyl covered floors (H_W)	0.97

Table J3D11j: Conductance and radiation factors: climate zone 4

Type of factor	Factor
Bedroom conduction weighting factor (BC)	0.70
Room type multiplier (for bedroom and unconditioned areas) (R_W)	1.00
Bedroom solar heat gain weighting factor (BS_W)	0.60
Floor factor for tiled or vinyl covered floors (H_W)	1.04
Floor factor for other than tiled or vinyl covered floors (H_W)	0.92

Table J3D11k: Conductance and radiation factors: climate zone 5

Type of factor	Factor
Bedroom conduction weighting factor (BC)	0.63
Room type multiplier (for bedroom and unconditioned areas) (R_W)	1.10
Bedroom solar heat gain weighting factor (BS_W)	0.81
Floor factor for tiled or vinyl covered floors (H_W)	1.03
Floor factor for other than tiled or vinyl covered floors (H_W)	0.93

Table J3D11l: Conductance and radiation factors: climate zone 6

Type of factor	Factor
Bedroom conduction weighting factor (BC)	0.81
Room type multiplier (for bedroom and unconditioned areas) (R_W)	1.00
Bedroom solar heat gain weighting factor (BS_W)	0.65
Floor factor for tiled or vinyl covered floors (H_W)	1.02
Floor factor for other than tiled or vinyl covered floors (H_W)	0.98

Table J3D11m: Conductance and radiation factors: climate zones 7 and 8

Type of factor	Factor
Bedroom conduction weighting factor (BC)	0.60
Room type multiplier (for bedroom and unconditioned areas) (R_W)	1.00
Bedroom solar heat gain weighting factor (BS_W)	0.51
Floor factor for tiled or vinyl covered floors (H_W)	1.03
Floor factor for other than tiled or vinyl covered floors (H_W)	0.93

Table J3D11n: Orientation sector conductance factor (OC)

Climate zone	North	North east	East	South east	South	South west	West	North west
2	1.70	1.34	0.98	0.84	0.70	0.90	1.10	1.40
3	1.30	1.10	0.90	0.95	1.00	0.95	0.90	1.10
4	1.30	1.25	1.20	1.03	0.85	0.92	0.99	1.15
5	1.20	1.15	1.10	1.05	1.00	1.05	1.10	1.15
6	1.23	1.13	1.00	1.00	1.02	1.00	1.00	1.16
7 and 8	1.40	1.25	1.10	1.00	0.90	0.95	1.00	1.20

Table J3D11o: Frame factor (F_W) — climate zone 2

Frame solar absorptance	Factor
≤ 0.40	0.97
> 0.40 to < 0.68	1.00
≥ 0.68	1.08

Table Notes

Interpolation is allowed for values between those shown.

Table J3D11p: Frame factor (F_W) — climate zone 3

Frame solar absorptance	Factor
≤ 0.40	0.98
> 0.40 to < 0.68	1.00
≥ 0.68	1.05

Table Notes

Interpolation is allowed for values between those shown.

Table J3D11q: Frame factor (F_W) — climate zone 4

Frame solar absorptance	Factor
≤ 0.40	0.99
> 0.40 to < 0.68	1.00
≥ 0.68	1.01

Table Notes

Interpolation is allowed for values between those shown.

Table J3D11r: Frame factor (F_W) — climate zone 5

Frame solar absorptance	Factor
≤ 0.40	1.00
> 0.40 to < 0.68	1.00
≥ 0.68	1.01

Table Notes

Interpolation is allowed for values between those shown.

Table J3D11s: Frame factor (F_W) — climate zone 6

Frame solar absorptance	Factor
≤ 0.40	0.92
> 0.40 to < 0.68	1.00
≥ 0.68	1.13

Table Notes

Interpolation is allowed for values between those shown.

Table J3D11t: Frame factor (F_W) — climate zones 7 and 8

Frame solar absorptance	Factor
≤ 0.40	0.97
> 0.40 to < 0.68	1.00
≥ 0.68	1.01

Table Notes

Interpolation is allowed for values between those shown.

NSW J3D12

J3D12 External summer glazing of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[New for 2022]

(1) In *climate zones* 1 to 7, the aggregate solar heat gain of the *glazing* in each *storey*, including any *mezzanine*, of a *sole-occupancy unit* of a Class 2 building or a Class 4 part of a building must—

(a) not exceed the allowance resulting from multiplying the *floor area* of the *storey*, including any *mezzanine*, measured within the enclosing walls, by the constant C_{SHGC} obtained from Table J3D12a; and

(b) be calculated in accordance with the following formula:

$$(A_1 \times SHGC_1 \times E_{S1} \times R_{S1} \times F_{S1} \times H_{S1}) + (A_2 \times SHGC_2 \times E_{S2} \times R_{S2} \times F_{S2} \times H_{S2}) + \ldots$$

(2) In the formula at (1)(b)—

(a) $A_{1,2,etc}$ = the area of each *glazing* element; and

(b) $SHGC_{1,2,etc}$ = the *Total System SHGC* for each *glazing* element, not exceeding 0.7; and

(c) $E_{S1,2,etc}$ = the summer exposure factor for each *glazing* element obtained from Table J3D12b, J3D12c, J3D12d, J3D12e, J3D12f, J3D12g or J3D12h.

(d) $R_{S1,2,etc}$ = the factor obtained from Table J3D12i or Table J3D12j for each *glazing* element located in a bedroom or room which is not a *conditioned space*; and

(e) $F_{S1,2,etc}$ = the frame factor obtained from Table J3D12i or Table J3D12j for *glazing* element; and

(f) $H_{S1,2,etc}$ = the floor factor obtained from Table J3D12i or Table J3D12j for each *glazing* element.

(3) For the purpose of J3D12—

(a) orientation sectors must be determined in accordance with Figure 13.3.2a of the ABCB Housing Provisions; and

(b) P/H must be determined in accordance with Figure S37C7; and

(c) for P/H between those shown in Tables J3D12b, J3D12c, J3D12d, J3D12e, J3D12f, J3D12g, J3D12h, J3D12i and J3D12j, either use the next highest P/H or interpolate.

Table J3D12a: Constant for Solar Heat Gain (C_{SHGC}) — climate zones 1 to 7

% Ventilation opening area per m²	*Climate zone* 1	*Climate zone* 2	*Climate zone* 3	*Climate zone* 4	*Climate zone* 5	*Climate zone* 6	*Climate zone* 7
5% to <10%	0.0191	0.0245	0.0547	0.0506	0.0674	0.1472	0.0930
10% to <15%	0.0237	0.0532	0.0745	0.0946	0.1111	0.2969	0.2405
15% to <20%	0.0294	0.0700	0.0861	0.1203	0.1367	0.3845	0.3267
20% to 90%	0.0364	0.0819	0.0943	0.1385	0.1548	0.4466	0.3879

Table Notes

(1) The *ventilation opening* area is the total area of each *ventilation opening* divided by the *floor area* of the *storey*, including any *mezzanine*.

(2) No window can have a design *ventilation opening* greater than 90% because the window frame will always obstruct some area of the opening.

(3) Interpolation is allowed for values between those shown.

Table J3D12b: Orientation sector summer exposure factor (E_s) — climate zone 1

P/H	North	North east	East	South east	South	South west	West	North west
0.00	0.62	0.76	0.79	0.75	0.52	0.89	1.06	0.85
0.05	0.52	0.67	0.73	0.68	0.45	0.82	0.98	0.75
0.10	0.49	0.61	0.68	0.65	0.41	0.76	0.92	0.69
0.20	0.44	0.53	0.62	0.57	0.33	0.66	0.80	0.60
0.40	0.36	0.41	0.48	0.42	0.25	0.53	0.66	0.47
0.60	0.30	0.33	0.40	0.36	0.20	0.42	0.53	0.38
0.80	0.26	0.28	0.32	0.28	0.18	0.36	0.46	0.32
1.00	0.23	0.23	0.27	0.25	0.15	0.30	0.38	0.28
1.20	0.21	0.21	0.23	0.20	0.13	0.27	0.34	0.26
1.40	0.20	0.19	0.20	0.18	0.13	0.22	0.30	0.22
1.60	0.18	0.16	0.17	0.16	0.11	0.20	0.27	0.21
1.80	0.15	0.16	0.16	0.13	0.10	0.20	0.24	0.18
2.00	0.14	0.15	0.14	0.13	0.10	0.17	0.22	0.17

Table J3D12c: Orientation sector summer exposure factor (E_s) — climate zone 2

P/H	North	North east	East	South east	South	South west	West	North west
0.00	0.65	1.16	1.59	1.20	0.73	1.30	1.77	1.23
0.05	0.54	1.01	1.43	1.06	0.61	1.18	1.62	1.10
0.10	0.50	0.94	1.35	0.99	0.58	1.10	1.50	1.00
0.20	0.42	0.81	1.20	0.88	0.51	0.98	1.32	0.87
0.40	0.35	0.62	0.95	0.70	0.40	0.78	1.10	0.67
0.60	0.30	0.48	0.78	0.57	0.33	0.64	0.88	0.50
0.80	0.26	0.41	0.65	0.47	0.29	0.54	0.77	0.43
1.00	0.23	0.33	0.56	0.40	0.24	0.47	0.62	0.36
1.20	0.21	0.30	0.46	0.35	0.22	0.40	0.54	0.31
1.40	0.19	0.26	0.42	0.32	0.21	0.35	0.48	0.27
1.60	0.17	0.25	0.36	0.29	0.19	0.31	0.41	0.24
1.80	0.15	0.22	0.31	0.25	0.17	0.30	0.37	0.22
2.00	0.15	0.21	0.29	0.24	0.16	0.26	0.36	0.21

Table J3D12d: Orientation sector summer exposure factor (E_s) — climate zone 3

P/H	North	North east	East	South east	South	South west	West	North west
0.00	0.80	1.26	1.41	1.38	0.89	1.33	1.29	1.20
0.05	0.67	1.14	1.31	1.26	0.77	1.21	1.20	1.07
0.10	0.63	1.03	1.24	1.19	0.73	1.14	1.13	0.99
0.20	0.54	0.88	1.09	1.05	0.62	1.00	1.01	0.87
0.40	0.46	0.68	0.87	0.83	0.51	0.83	0.80	0.67
0.60	0.40	0.52	0.73	0.68	0.42	0.66	0.67	0.52
0.80	0.34	0.42	0.58	0.55	0.36	0.58	0.57	0.42
1.00	0.29	0.35	0.50	0.47	0.32	0.49	0.50	0.35
1.20	0.27	0.31	0.42	0.40	0.28	0.43	0.41	0.31
1.40	0.24	0.27	0.35	0.36	0.27	0.37	0.39	0.27

P/H	North	North east	East	South east	South	South west	West	North west
1.60	0.24	0.24	0.33	0.32	0.22	0.36	0.33	0.25
1.80	0.21	0.23	0.30	0.28	0.22	0.32	0.31	0.23
2.00	0.21	0.22	0.25	0.28	0.20	0.28	0.26	0.20

Table J3D12e: Orientation sector summer exposure factor (E_s) — climate zone 4

P/H	North	North east	East	South east	South	South west	West	North west
0.00	0.79	1.13	1.12	1.05	0.68	1.14	1.44	1.23
0.05	0.67	1.05	1.05	0.97	0.59	1.05	1.34	1.12
0.10	0.62	0.95	0.99	0.91	0.55	0.98	1.29	1.03
0.20	0.47	0.83	0.90	0.82	0.49	0.87	1.16	0.89
0.40	0.33	0.63	0.74	0.67	0.41	0.71	0.94	0.68
0.60	0.30	0.48	0.59	0.56	0.35	0.61	0.79	0.50
0.80	0.26	0.36	0.50	0.49	0.30	0.52	0.65	0.40
1.00	0.22	0.29	0.44	0.42	0.26	0.45	0.56	0.33
1.20	0.21	0.25	0.37	0.37	0.24	0.40	0.50	0.28
1.40	0.18	0.22	0.31	0.34	0.20	0.38	0.42	0.23
1.60	0.18	0.19	0.30	0.30	0.19	0.33	0.36	0.22
1.80	0.15	0.17	0.26	0.27	0.17	0.29	0.35	0.19
2.00	0.14	0.16	0.22	0.23	0.17	0.28	0.29	0.18

Table J3D12f: Orientation sector summer exposure factor (E_s) — climate zone 5

P/H	North	North east	East	South east	South	South west	West	North west
0.00	0.82	1.20	1.31	1.06	0.82	1.04	1.30	1.16
0.05	0.69	1.06	1.18	0.94	0.68	0.92	1.19	1.04
0.10	0.63	0.97	1.11	0.87	0.65	0.86	1.11	0.94
0.20	0.51	0.84	0.98	0.77	0.58	0.76	0.99	0.83
0.40	0.39	0.64	0.78	0.63	0.46	0.62	0.81	0.62
0.60	0.35	0.51	0.64	0.52	0.40	0.51	0.65	0.48
0.80	0.30	0.41	0.55	0.44	0.34	0.43	0.52	0.40
1.00	0.26	0.34	0.46	0.37	0.30	0.37	0.46	0.31
1.20	0.24	0.29	0.40	0.33	0.26	0.33	0.40	0.27
1.40	0.21	0.25	0.35	0.30	0.24	0.29	0.34	0.24
1.60	0.20	0.24	0.32	0.25	0.22	0.27	0.30	0.21
1.80	0.18	0.22	0.28	0.23	0.20	0.23	0.27	0.20
2.00	0.17	0.19	0.26	0.23	0.19	0.21	0.25	0.19

Table J3D12g: Orientation sector summer exposure factor (E_s) — climate zone 6

P/H	North	North east	East	South east	South	South west	West	North west
0.00	2.18	2.75	2.88	2.22	1.59	2.46	2.91	2.90
0.05	1.85	2.47	2.63	1.99	1.35	2.25	2.70	2.64
0.10	1.69	2.30	2.48	1.89	1.27	2.13	2.60	2.43
0.20	1.35	1.96	2.20	1.66	1.14	1.92	2.33	2.13
0.40	0.94	1.48	1.78	1.38	0.94	1.57	1.87	1.61

P/H	North	North east	East	South east	South	South west	West	North west
0.60	0.78	1.10	1.53	1.15	0.81	1.36	1.58	1.19
0.80	0.68	0.89	1.25	0.97	0.68	1.17	1.37	0.94
1.00	0.57	0.74	1.05	0.82	0.60	0.98	1.16	0.84
1.20	0.52	0.61	0.93	0.74	0.60	0.91	1.00	0.68
1.40	0.47	0.56	0.80	0.66	0.49	0.80	0.87	0.61
1.60	0.42	0.48	0.70	0.61	0.47	0.73	0.79	0.49
1.80	0.39	0.46	0.65	0.56	0.44	0.66	0.71	0.47
2.00	0.36	0.43	0.60	0.54	0.44	0.61	0.64	0.40

Table J3D12h: Orientation sector summer exposure factor (E_s) — climate zone 7

P/H	North	North east	East	South east	South	South west	West	North west
0.00	0.89	1.06	1.06	0.93	0.70	0.91	1.06	1.07
0.05	0.77	0.95	0.97	0.82	0.59	0.81	0.97	0.97
0.10	0.71	0.88	0.92	0.79	0.56	0.76	0.92	0.89
0.20	0.58	0.77	0.82	0.69	0.50	0.68	0.81	0.78
0.40	0.37	0.59	0.67	0.57	0.42	0.55	0.66	0.58
0.60	0.30	0.46	0.57	0.50	0.36	0.47	0.56	0.46
0.80	0.26	0.36	0.48	0.44	0.31	0.41	0.47	0.36
1.00	0.23	0.30	0.42	0.37	0.28	0.35	0.39	0.29
1.20	0.20	0.25	0.36	0.34	0.25	0.31	0.34	0.25
1.40	0.18	0.21	0.32	0.30	0.23	0.28	0.29	0.22
1.60	0.17	0.19	0.29	0.27	0.22	0.26	0.28	0.19
1.80	0.16	0.18	0.25	0.24	0.20	0.24	0.24	0.17
2.00	0.15	0.17	0.24	0.23	0.20	0.21	0.22	0.16

Table J3D12i: Conductance factors — climate zone 1

Factor	Value
Room type multiplier (for bedroom and unconditioned areas) (R_S)	0.32
Frame solar absorptance multiplier Frame SA = 0.3 up to 0.4 (F_S)	0.83
Frame solar absorptance multiplier Frame SA = 0.4 up to 0.5 (F_S)	1.00
Frame solar absorptance multiplier Frame SA = 0.5 or more (F_S)	1.20
Floor factor for tiled or vinyl covered floors (H_S)	1.00
Floor factor for other than tiled or vinyl covered floors (H_S)	1.00

Table J3D12j: Conductance factors — climate zones 2 to 7

Type of factor	*Climate zone* 2	*Climate zone* 3	*Climate zone* 4	*Climate zone* 5	*Climate zone* 6	*Climate zone* 7
Room type multiplier (for bedroom and unconditioned areas) (R_S)	0.40	0.56	0.71	0.91	0.87	1.11
Frame solar absorptance multiplier (for metal frame windows) (F_S) SA ≤ 0.40	1.00	0.89	0.87	0.85	0.74	0.86
Frame solar absorptance multiplier (for metal frame windows) (F_S) SA > 0.40 to < 0.68	1.06	1.00	1.00	1.00	1.00	1.00
Frame solar absorptance multiplier (for metal frame windows) (F_S) SA ≥ 0.68	1.22	1.18	1.22	1.24	1.22	1.32
Floor factor for tiled or vinyl covered floors (H_S)	1.06	1.06	1.13	1.13	1.04	1.21
Floor factor for other than tiled or vinyl covered floors (H_S)	0.97	0.97	0.94	0.94	0.94	0.90

Table Notes

Interpolation is allowed for values between those shown.

NSW J3D13

J3D13 Shading of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[New for 2022]

Where shading is *required* to comply with J3D11 or J3D12, it must—

(a) be provided by an external permanent projection, such as a verandah, balcony, fixed canopy, eaves, shading hood or carport, which—

(i) extends horizontally on both sides of the *glazing* for a distance greater than or equal to the projection distance P in Figure S37C7; or

(ii) provide the equivalent shading to (i) with a reveal or the like; or

(b) be provided by an external shading device, such as a shutter, blind, vertical or horizontal building screen with blades, battens or slats, which—

(i) is capable of restricting at least 80% of the summer solar radiation; and

(ii) if adjustable, is readily operated either manually, mechanically or electronically by the building occupants.

NSW J3D14

J3D14 Net equivalent energy usage of a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

[New for 2022]

(1) The net equivalent energy usage of a *sole-occupancy unit* of a Class 2 building or Class 4 part of a building, calculated in accordance with (a), must not exceed the allowance calculated in accordance with (b)—

(a) $(A \times E_E) + E_P + E_S - E_R$, where—

(i) A = the floor area factor obtained from multiplying the total floor area by the adjustment factor in Table J3D14a; and

(ii) E_E = the *main space conditioning* and *main water heater* efficiency factor obtained from the ABCB Standard for Whole-of-Home Efficiency Factors; and

(iii) E_P = the swimming pool pump energy usage in (2); and

(iv) E_S = the spa pump energy usage in (3); and

(v) E_R = the installed capacity of on-site photovoltaics apportioned to the *sole-occupancy unit* of a Class 2 building or Class 4 part of a building (kW); and

(b) $A \times E_F$, where—

(i) A = the floor area factor obtained from multiplying the total floor area by the adjustment factor in Table J3D14a; and

(ii) E_F = the energy factor obtained from Table J3D14b.

(2) The swimming pool pump energy usage (E_P) must be determined in accordance with the following formula: $E_P = V \times FP / 1000$, where—

(a) E_P = the swimming pool pump energy usage; and

(b) V = the volume of the swimming pool to the nearest 1000 litres; and

(c) FP = the swimming pool pump factor in Table 13.6.2c of the ABCB Housing Provisions.

(3) The spa pump energy usage (E_S) must be determined in accordance with the following formula: $E_S = V \times FSB / 100$,where—

(a) ES = the spa pump energy usage; and

(b) V = the volume of the spa to the nearest 100 litres; and

(c) FSB = the spa pump factor in Table 13.6.2d of the ABCB Housing Provisions.

Table J3D14a: Floor area adjustment factor for a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

Total floor area m^2	Floor area factor	Total floor area m^2	Floor area factor	Total floor area m^2	Floor area factor	Total floor area m^2	Floor area factor
< 50	0.0123	160–169	0.0097	280–289	0.0087	400–409	0.0080
50–59	0.0119	170–179	0.0096	290–299	0.0086	410–419	0.0079
60–69	0.0116	180–189	0.0095	300–309	0.0085	420–429	0.0079
70–79	0.0113	190–199	0.0094	310–319	0.0085	430–439	0.0078

Total floor area m^2	Floor area factor	Total floor area m^2	Floor area factor	Total floor area m^2	Floor area factor	Total floor area m^2	Floor area factor
80–89	0.0111	200–209	0.0093	320–329	0.0084	440–449	0.0078
90–99	0.0108	210–219	0.0092	330–339	0.0083	450–459	0.0077
100–109	0.0106	220–229	0.0091	340–349	0.0083	460–469	0.0077
110–119	0.0105	230–239	0.0090	350–359	0.0082	470–479	0.0077
120–129	0.0103	240–249	0.0090	360–369	0.0082	480–489	0.0076
130–139	0.0101	250–259	0.0089	370–379	0.0081	490–499	0.0076
140–149	0.0100	260–269	0.0088	380–389	0.0081	500	0.0075
150–159	0.0099	270–279	0.0087	390–399	0.0080	—	—

Table Notes

(1) The total floor area is measured within the inside face of the *external walls* of the *sole-occupancy unit* and includes any conditioned attached Class 10a building.

(2) Where values fall between ranges given, the floor area must be rounded up to the nearest whole square metres of floor area.

Table J3D14b: Energy factor for a sole-occupancy unit of a Class 2 building or a Class 4 part of a building

Climate zone	ACT	NSW	NT	QLD	SA	TAS	Vic	WA
1	—	—	2.73	3.95	—	—	—	4.64
2	—	1.88	—	2.54	—	—	—	—
3	—	—	1.76	3.52	—	—	—	4.10
4	—	2.57	—	—	2.65	—	1.79	3.34
5	—	2.50	—	3.26	2.56	—	—	3.36
6	—	3.43	—	—	3.58	—	2.32	4.58
7	3.66	3.32	—	—	—	4.41	2.32	—
8	—	5.70	—	—	—	5.60	4.02	—

NSW J3D15

J3D15 Net equivalent energy usage for a sole-occupancy unit of a Class 2 building or Class 4 part of building – home energy rating software

[New for 2022]

A *sole-occupancy unit* of a Class 2 building or a Class 4 part of a building must achieve a whole-of-home rating of not less than 50 using *house energy rating software*.